

Table 1: Dataset structural properties, verified via `dataset_properties_report.py`. Sample counts are number of trials per split. [†] sampling rate assumed from BIOT config, not independently re-verified.

Domain	Dataset	Ch.	Rate (Hz)	Trial (s)	#Cls	Train 100%	Train 50%	Train 25%	Val	Test
Clinical	CHB-MIT	16	256	10.000	2	28800	14400	7200	23065	16964
Clinical	Alzheimer’s	19	500	10.000	2	563	281	141	605	2241
Sleep	ISRUC	6	200	30.000	5	9600	4800	2400	9360	8700
Sleep	DREAM	257	500	20.000	2	120	60	30	42	43
Speech	Karaone	60	1000	4.964	2	956	478	239	396	561
Speech	JapanEEG	128	200	10.000	2	28798	14398	7199	4728	4928
Visual	Yang2025	59	200 [†]	7.500	2	24598	12299	6149	3000	2993
Visual	ThoughtViz	14	200 [†]	0.160	10	1000	500	250	500	5706

Table 2: Dataset recording duration (train 100% + val + test), and train-only duration at 100% data.

Domain	Dataset	Total dur. (h)	Train-only dur., 100% (h)
Clinical	CHB-MIT	191.19	80.00
Clinical	Alzheimer’s	9.47	1.56
Sleep	ISRUC	230.50	80.00
Sleep	DREAM	1.14	0.67
Speech	Karaone	2.64	1.32
Speech	JapanEEG	106.82	79.99
Visual	Yang2025	63.73	51.25
Visual	ThoughtViz	0.32	0.04

Table 3: Class balance at 100% training data: majority vs. minority class share of the train(100%) split.

Domain	Dataset	Majority class %	Minority class %	Note
Clinical	CHB-MIT	99.3%	0.7%	severe imbalance
Clinical	Alzheimer’s	57.9%	42.1%	
Sleep	ISRUC	27.7%	12.3%	
Sleep	DREAM	54.2%	45.8%	
Speech	Karaone	81.9%	18.1%	
Speech	JapanEEG	77.6%	22.4%	
Visual	Yang2025	50.0%	50.0%	balanced
Visual	ThoughtViz	10.0%	10.0%	balanced, 10-way